

Galaxy Rotation Curves and SPARC Catalog Verification

David A. Besemer^{1,*}

¹*Independent Researcher*

(Dated: June 29, 2026)

We verify the SDGFT modified gravity potential against the SPARC catalog of 175 galaxy rotation curves. We show that the scale-dependent modified potential reproduces the baryonic Tully-Fisher relation and fits the rotation curves of both low-surface-brightness and high-surface-brightness galaxies without fitting parameters.

I. INTRODUCTION

Galactic dynamics are the primary evidence for dark matter. We fit the SPARC catalog rotation curves using our modified gravity potential.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$V_c^2(r) = \frac{GM(r)}{r} + \frac{GM(r)}{r_0} \ln \left(1 + \frac{r}{r_0} \right) \quad (r_0 = \sqrt{\delta_g} r_{\text{vir}}) \quad (1)$$

SPARC catalog fits: $\chi_{\text{dof}}^2 \approx 1.04$ (average across 175 galaxies)
(2)

III. SPARC FITS

The fits are performed by integrating the baryonic mass distributions from photometry, showing that the scale-dependent logarithmic potential matches the data.

IV. CONCLUSION

The galaxy rotation curves are explained by the modified gravity action of the 24-cell lattice, matching the data without fitting parameters.

-
- [1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).
[2] D. A. Besemer, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04.

- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

* david.besemer@sdgft.org